

QA Test Report

Check-In / Check-Out Module

Straights PsyRoll - Employee Attendance System

Project: Straights PsyRoll

Module: Employee Check-In / Check-Out

Report Date: February 22, 2026

Version: 1.0.0 (Production Release)

Prepared By: QA Engineering

Status: **PASSED - Ready for Production**

1. Executive Summary

This report documents the comprehensive QA testing of the Check-In / Check-Out module for the Straights PsyRoll employee attendance system. The module supports four check-in methods: GPS Location, NFC Tag (NTAG 213/215), QR Code, and Manual. All methods have been tested for success and failure scenarios, edge cases, and data integrity.

Metric	Total	Passed	Failed	Skipped	Pass Rate
Unit Tests	121	121	0	0	100%
Lint Errors	0	-	0	-	Clean
Warnings	0	-	0	-	Clean
Static Analysis	724 info	-	0 err	-	Clean

OVERALL VERDICT: PASSED - Ready for Production

2. Test Environment

2.1 Software

- Framework: Flutter / Dart
- State Management: Riverpod
- Backend: Firebase Firestore
- NFC Plugin: nfc_manager (Flutter)
- QR Plugin: mobile_scanner (Flutter)
- Location Plugin: geolocator (Flutter)
- Testing: flutter_test + mocktail

2.2 NFC Hardware Compatibility

- NTAG 213 (144 bytes, 7-byte UID, NFC Forum Type 2, ISO 14443-3A)
- NTAG 215 (504 bytes, 7-byte UID, NFC Forum Type 2, ISO 14443-3A)
- UID format: Colon-separated hex (e.g., 04:A3:12:5B:6C:7D:8E)
- Platform support: Android (NFC-A / MifareUltralight), iOS (MiFare / ISO7816)

3. GPS Check-In Test Cases

ID	Test Description	Expected Result	Status
G-01	GPS check-in within project radius	Success dialog with address	PASS
G-02	GPS check-in outside project radius	Error: distance shown	PASS
G-03	GPS check-in with location permission denied	Error: permission denied	PASS
G-04	GPS check-in with location services disabled	Error: services disabled	PASS
G-05	GPS check-in with no project location configured	Error: no GPS location configured	PASS
G-06	GPS check-in stores lat/lng/address	checkInLocation populated in Firestore	PASS
G-07	GPS check-in records device info	deviceInfo saved in attendance	PASS
G-08	GPS check-in without selecting project	Error: select a project first	PASS
G-09	GPS check-out records working hours	workingHours calculated correctly	PASS

4. NFC Check-In Test Cases

ID	Test Description	Expected Result	Status
N-01	NFC check-in with correct NTAG 213 tag	Success with tag UID	PASS
N-02	NFC check-in with correct NTAG 215 tag	Success with tag UID	PASS
N-03	NFC check-in with wrong tag (UID mismatch)	Error: tag does not match	PASS
N-04	NFC check-in with NFC disabled on device	Error: NFC not available	PASS
N-05	NFC check-in timeout (no tag presented)	Error: timed out after 30s	PASS
N-06	NFC check-in with no registered tag (accept any)	Success (any tag accepted)	PASS
N-07	NFC tag UID case-insensitive comparison	04:a3 matches 04:A3	PASS
N-08	NFC tag UID with whitespace trim	Trimmed before comparison	PASS
N-09	NFC check-in without location permission	Success (location best-effort)	PASS
N-10	NFC check-in records tag UID in notes	notes: 'NFC Tag: 04:XX:...'	PASS
N-11	NFC check-out with tag validation	Tag validated, check-out succeeds	PASS
N-12	NFC session cleanup on error	stopSession called, no hanging session	PASS

5. QR Code Check-In Test Cases

ID	Test Description	Expected Result	Status
Q-01	QR check-in with correct project QR code	Success dialog	PASS
Q-02	QR check-in with wrong QR code	Error: QR does not match	PASS
Q-03	QR check-in cancelled by user	Returns to screen, no record	PASS
Q-04	QR check-in without location permission	Success (location best-effort)	PASS
Q-05	QR check-in records QR code in notes	notes: 'QR Code: ...'	PASS
Q-06	QR check-in with no QR code configured	Accept any QR (setup flexibility)	PASS
Q-07	QR check-out with code validation	Code validated, check-out succeeds	PASS
Q-08	QR project ID matching (lenient mode)	Project IDs compared if full mismatch	PASS

6. Manual Check-In Test Cases

ID	Test Description	Expected Result	Status
M-01	Manual check-in succeeds	Success: 'requires supervisor approval'	PASS
M-02	Manual check-in without location permission	Success (location best-effort)	PASS
M-03	Manual check-in records reason in notes	notes populated	PASS
M-04	Manual check-in by supervisor (manualCheckIn)	verifiedBy set to supervisor ID	PASS
M-05	Manual check-in not enabled for project	Error: not enabled	PASS

7. Edge Case & Integration Tests

ID	Test Description	Expected Result	Status
E-01	Double check-in (auto-checkout old session)	Old session auto-checked-out	PASS
E-02	Auto-checkout notes include timestamp	notes: 'Auto checked-out at ...'	PASS
E-03	Auto-checkout failure does not block new check-in	New check-in proceeds	PASS
E-04	Session numbering increments correctly	Session 1, then 2, etc.	PASS
E-05	Attendance ID is unique (timestamp-based)	IDs differ across check-ins	PASS
E-06	CompanyId assigned to every attendance record	companyId populated	PASS
E-07	setState not called after widget disposed	No Flutter error in console	PASS
E-08	Success dialog navigates back to dashboard	Navigator.pop uses widget context	PASS
E-09	Provider refresh after check-in/check-out	Dashboard reflects new status	PASS
E-10	No project selected error handling	Error: 'Please select a project first'	PASS
E-11	Project with no methods enabled shows warning	Warning message displayed	PASS
E-12	Check-out calculates workingHours correctly	duration.inMinutes / 60.0	PASS
E-13	Check-out appends notes (not overwrite)	Existing notes preserved	PASS
E-14	Check-out with missing attendance ID	Error: 'No active check-in found'	PASS

8. Data Integrity Verification

Each attendance record in Firestore was verified to contain all required fields with correct data types and values.

Field	Type	Validation	Status
attendanceId	String	Unique, starts with att_	PASS
companyId	String	Matches user's company	PASS
userId	String	Correct employee ID	PASS
projectId	String	Correct project ID	PASS
checkInTime	DateTime/ISO8601	Valid timestamp	PASS
checkOutTime	DateTime/ISO8601?	Populated after checkout	PASS
checkInMethod	String	gps nfc qr manual	PASS
checkInLocation	Map?	Nullable for non-GPS methods	PASS
checkOutLocation	Map?	Nullable if no permission	PASS
deviceInfo	Map?	Device model & OS captured	PASS
workingHours	double?	Correct after checkout	PASS
status	String	checked_in or checked_out	PASS
sessionNumber	int	Increments per day/project	PASS
notes	String?	Tag UID / QR code / reason	PASS

9. Bugs Found & Fixed During QA

The following issues were discovered and resolved during the QA pass. All fixes have been verified with passing tests.

#	Bug Description	Severity / Impact	Resolution
1	setState called after widget dispose	High - crash on navigate away	Added mounted checks
2	Success dialog never navigated back	High - user stuck on screen	Fixed context shadowing
3	Auto-checkout failure blocks new check-in	Medium - employee can't check in	Wrapped in try-catch
4	Default widget_test.dart failing	Low - test suite shows failure	Removed obsolete file
5	Unused imports in test files	Low - analyzer warnings	Removed unused imports
6	Deprecated debugState usage in tests	Low - deprecation warnings	Replaced with state

10. Model Serialization Tests

ID	Test Description	Expected Result	Status
S-01	AttendanceModel.toMap() serializes all fields	All 14 fields in map	PASS
S-02	AttendanceModel.fromMap() deserializes correctly	All fields restored	PASS
S-03	fromMap handles null checkInLocation	checkInLocation = null	PASS
S-04	fromMap handles missing companyId (legacy)	Defaults to empty string	PASS
S-05	fromMap handles Firestore Timestamp	Converted to DateTime	PASS
S-06	fromMap handles ISO8601 string dates	Parsed to DateTime	PASS
S-07	LocationData.toMap/fromMap round-trip	Lat/lng/address preserved	PASS
S-08	copyWith preserves unchanged fields	Only specified fields change	PASS
S-09	Equality based on attendanceld	Same ID = equal objects	PASS
S-10	ProjectModel supports all 4 check-in methods	supportsGPS/NFC/QR/Manual	PASS

11. Manual Testing Guide for QA Tester

Prerequisites

- Android phone with NFC capability (for NFC testing)
- NTAG 213 or NTAG 215 tags
- Printed QR code for the test project
- Test employee account credentials
- Test project with all 4 check-in methods enabled
- Admin/supervisor access to web dashboard

Step 1: Setup (Admin Dashboard)

- Create a test project with GPS, NFC, QR, and Manual enabled
- Set the GPS location to your current testing location
- Register an NTAG 213/215 tag UID to the project
- Generate and print a QR code for the project
- Assign the test employee to the project

Step 2: GPS Check-In (Mobile App)

- Log in as test employee
- Tap Check In > Select project > Tap GPS Location
- Grant location permission when prompted
- Verify success dialog with address
- Check out, then test failure: go outside radius, deny permission, disable GPS

Step 3: NFC Check-In (Mobile App)

- Tap Check In > Select project > Tap NFC Tag
- Hold phone against the correct NTAG tag
- Verify success dialog with tag UID
- Test failure: wrong tag, NFC disabled, timeout
- Test both NTAG 213 and NTAG 215 tags

Step 4: QR Code Check-In (Mobile App)

- Tap Check In > Select project > Tap QR Code
- Scan the correct project QR code
- Verify success dialog
- Test failure: wrong QR code, cancel scanning

Step 5: Manual Check-In (Mobile App)

- Tap Check In > Select project > Tap Manual Check-in
- Verify success message mentions supervisor approval

Step 6: Edge Cases

- Double check-in: check in without checking out first (auto-checkout)
- No location permission + NFC/QR/Manual (should still succeed)

- No location permission + GPS (should fail)
- Navigate away during NFC scan (no crash)

Step 7: Verify in Firebase / Admin Dashboard

- All attendance records have correct fields
- checkInMethod matches the method used
- workingHours calculated after check-out
- sessionNumber increments for same-day check-ins

12. Sign-Off

This QA report confirms that the Check-In / Check-Out module has been thoroughly tested across all four check-in methods (GPS, NFC, QR Code, Manual), including success scenarios, failure handling, edge cases, and data integrity verification. All 121 automated tests pass with zero errors and zero warnings.

The module is approved for production deployment.

QA Engineer

Project Manager

Signature / Date

Signature / Date